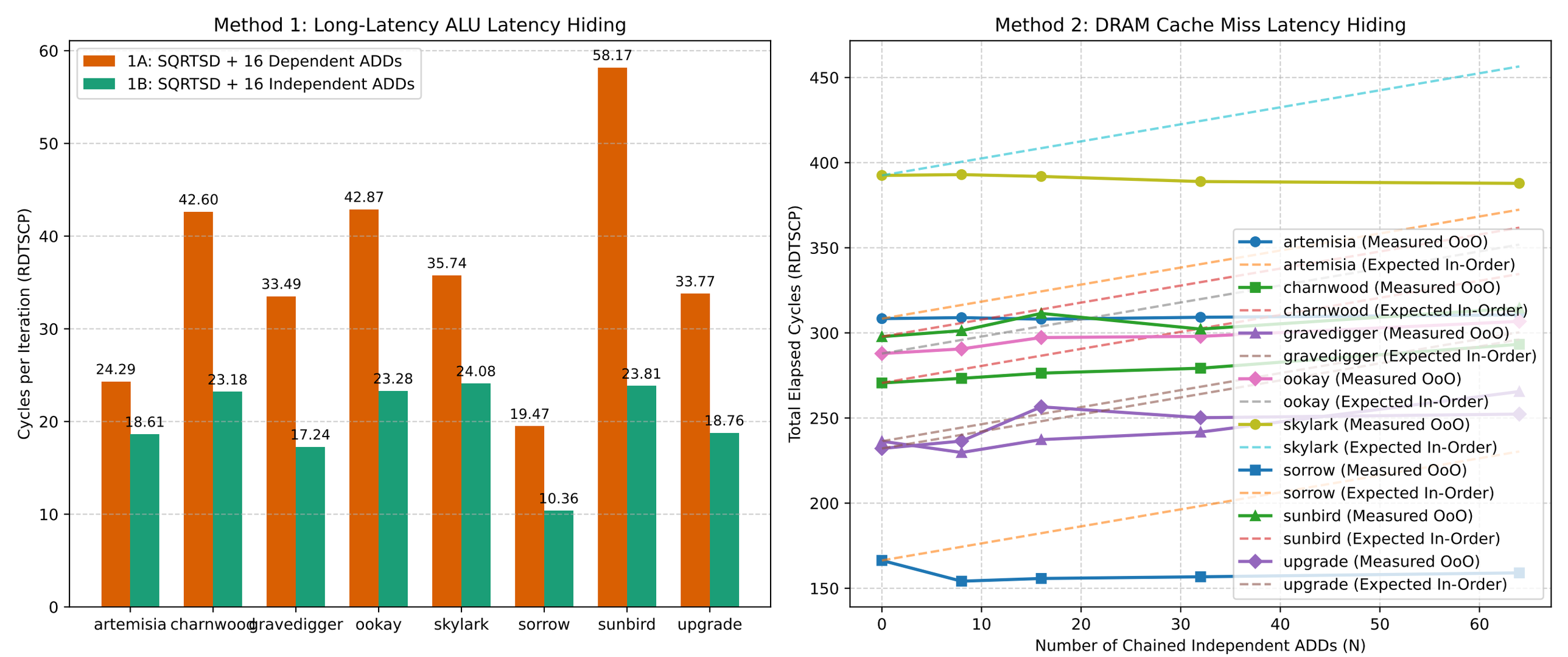
Method 1: Long-Latency ALU Latency Hiding
1A: SQRTSD + 16 Dependent ADDs
1B: SQRTSD + 16 Independent ADDs
Cycles per Iteration (RDTSCP)
0
10
20
30
40
50
60
24.29
18.61
42.60
23.18
33.49
17.24
42.87
23.28
35.74
24.08
19.47
10.36
58.17
23.81
33.77
18.76
artemisia
charnwood
gravedigger
ookay
skylark
sorrow
sunbird
upgrade
Method 2: DRAM Cache Miss Latency Hiding
Total Elapsed Cycles (RDTSCP)
150
200
250
300
350
400
450
0
10
20
30
40
50
60
Number of Chained Independent ADDs (N)
artemisia (Measured OoO)
artemisia (Expected In-Order)
charnwood (Measured OoO)
charnwood (Expected In-Order)
gravedigger (Measured OoO)
gravedigger (Expected In-Order)
ookay (Measured OoO)
ookay (Expected In-Order)
skylark (Measured OoO)
skylark (Expected In-Order)
sorrow (Measured OoO)
sorrow (Expected In-Order)
sunbird (Measured OoO)
sunbird (Expected In-Order)
upgrade (Measured OoO)
upgrade (Expected In-Order)